

The Top 10 Open Source Web Servers in 2019

A new year has begun and it's the right time to take stock of the software that was popular over the past year. So here is a list of the top ten open source Web servers in 2019.

A Web server can refer to hardware or software, or both working together.

- On the hardware side, a Web server is a computer that stores Web server software and a website's component files (e.g., HTML documents, images, CSS style sheets and JavaScript files). It is connected to the Internet, and supports physical data interchange with other devices connected to the Web.
- On the software side, a Web server includes several parts that control how Web users access hosted files; at the minimum, it is an HTTP server. An HTTP server is software that understands URLs (Web addresses) and HTTP (the protocol your browser uses to view Web pages). It can be accessed through the domain names (like *mozilla.org*) of the websites it stores, and it delivers their content to the end user's device.

At the most basic level, whenever a browser needs a file that is hosted on a Web server, it requests the file via HTTP. When the request reaches the correct Web server (hardware), the HTTP server (software) accepts the request, finds the requested document (if it doesn't, then a 404 response is returned), and sends it back to the browser, also through HTTP.

Publishing a website requires either a static or a dynamic Web server.



- A static Web server, or stack, consists of a computer (hardware) with an HTTP server (software). We call it static because the server sends its hosted files as is, to your browser.
- A dynamic Web server consists of a static Web server plus extra software, most commonly an application server and a database. We call it dynamic because the application server updates the hosted files before sending them to your browser via the HTTP server. The following are the main features of a Web server:
 - Creates one or more websites.
 - Configures log file settings, including where the log files are saved, what data is included in the log files, etc.
 - Configures website/directory security. For example, based on requirements and subsequent configuration, the Web server either allows or prevents a particular website being viewed, as well as which IP addresses are/aren't allowed to view the website, etc.
 - Creates an FTP site.
 - Creates virtual directories, and maps them to physical directories.
 - Configures/ nominates custom error pages. This allows the users to build and display user friendly error messages on the website.
 - Specifies default documents, which are displayed when no file name is given. For example, it specifies which files should be displayed if the user opens *http://localhost*. This is typically *index.html* or something similar, but it doesn't need to be.

Web servers have come a long way since the CERN *httpd* was developed by Tim Berners-Lee in 1990, as part of the project that resulted in the first Web browser. Some of the leading providers of Web servers provide closed source options for corporations, but many are still open source (based on Linux). Some Web servers are designed for certain special needs, some receive updates frequently, some are designed for certain technologies only, and some are known for their stability-cum-security because of the latest updates.